

Bitcoin & Quantum Computing

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Bitcoin & Quantum Computing

- ✦ Bitcoin threat model
- ✦ Current state
- ✦ Timeline for Solution



Quantum computing

- ✦ Leverages the principles of quantum mechanics
- ✦ Allows significant speed-ups in solving specific problems
- ✦ Cryptographically Relevant Quantum Computers ,CRQCs, could break ECC-based cryptography
- ✦ To date, no quantum computer has outperformed a supercomputer on any commercially relevant application

Quantum computing and Bitcoin

- ✧ Mining
 - ✧ Quantum miners running Grover's Algorithm efficiently
- ✧ Signatures
 - ✧ Cryptographically Relevant Quantum Computers ,CRQCs, breaking cryptographic assumptions (ECC)

Mining

Mining goal: Find a nonce that hashes below a target

Classical mining: More attempts = higher chance

- Linear growth, parallelizable

Quantum mining: Longer runtime = higher chance

- Square-root speedup, not parallelizable

Bitcoin Threat Model - Mining

- ✦ Mining will be significantly less affected in the foreseeable future
- ✦ If quantum mining will become realistic, we can expect:
 - ✦ Significant mining centralization
 - ✦ Increase in stale blocks
 - ✦ Weakening of security assumptions (51% attack)

Asymmetric Cryptography

- ✦ There are two keys - Public Key (**PK**) and Secret Key (**SK**)
- ✦ A **SK** can be used to sign a message
- ✦ The base assumption are:
 - ✦ Calculating **PK** from **SK** - **Easy**
 - ✦ Calculating **SK** from **PK** - **Infeasible**
 - ✦ Forging a signature having only **PK** - **Infeasible**

What would CRQC break?

- ✦ **EVERYTHING!** (Well, almost)
- ✦ Calculating **SK** from **PK** - **Easy**
- ✦ Forging a signature having only **PK** - **Easy**
- ✦ The attacker needs access to the **PK**
- ✦ Quantum secure signatures are known, but **lack efficiency or capabilities**
- ✦ There is **active ongoing research** in the cryptography community

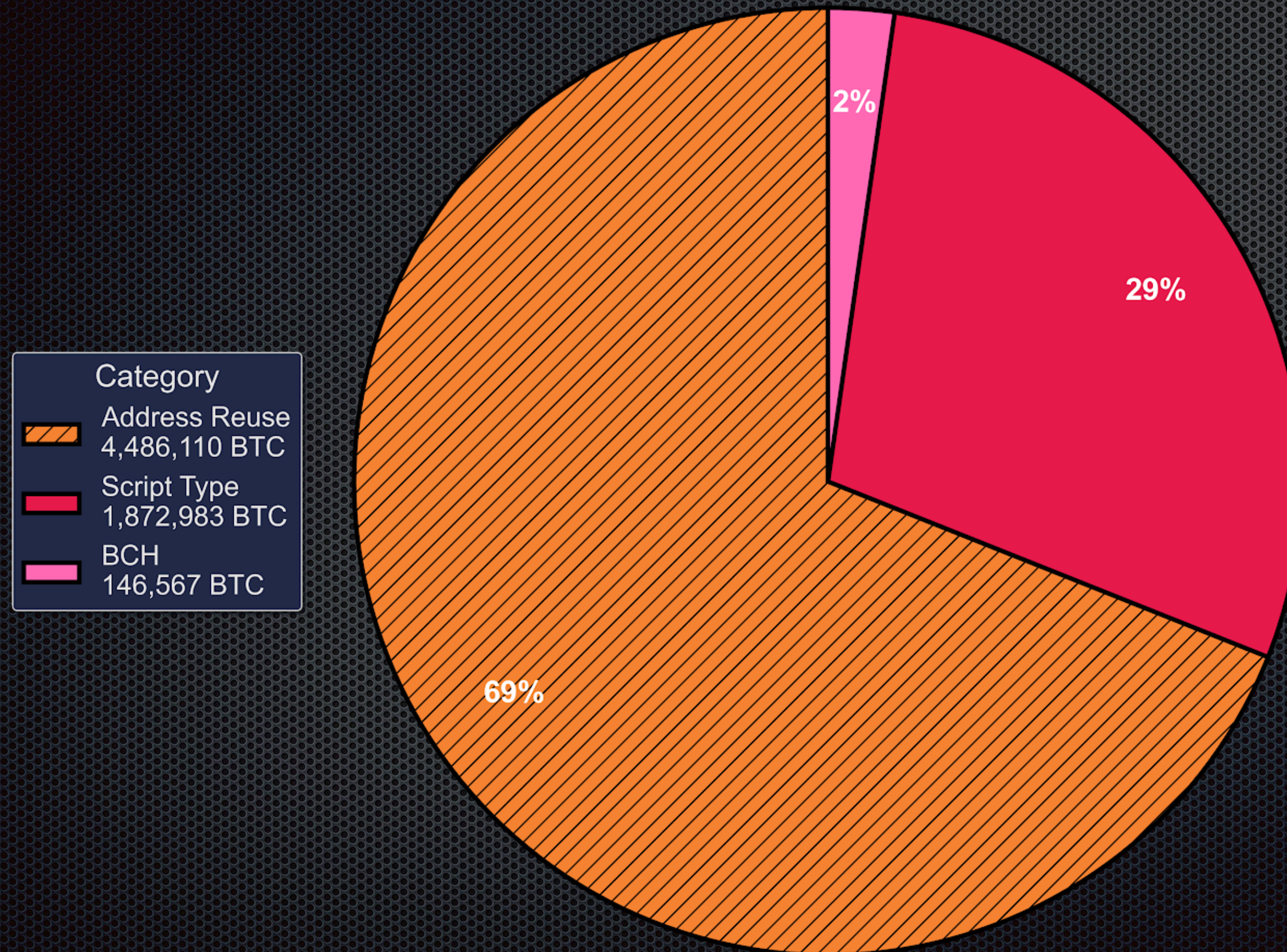
Cryptography and Bitcoin

- ✦ For a bitcoin transaction to be valid it needs to be signed by the **SK** matching the **PK** of the utxo
- ✦ In some cases (**P2PK**, **P2TR**) the **PK** is present on-chain
- ✦ In others (**P2WPKH**, **P2WSH**) a **hash** of the **PK** is on-chain, and needs to be revealed together with a signature to move the funds

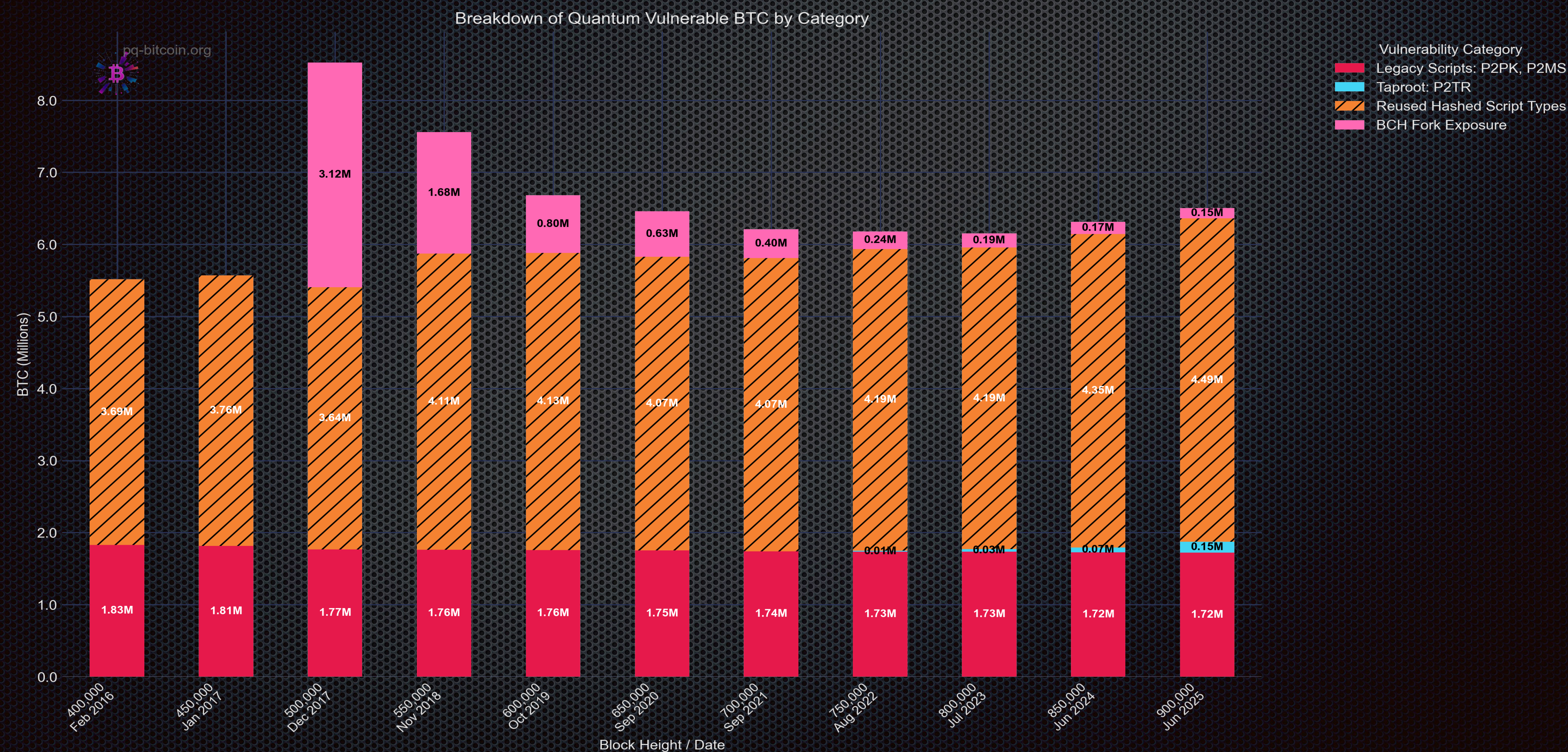
Bitcoin Threat Model - Signatures

- ✦ Some utxo are **immediately** vulnerable (long-range attack)
 - ✦ P2PK, P2MS, P2TR
 - ✦ Address re-use
 - ✦ BCH and other forks
- ✦ All other will be vulnerable **upon spend** (short-range attack)

Immediately vulnerable ~6.51M BTC



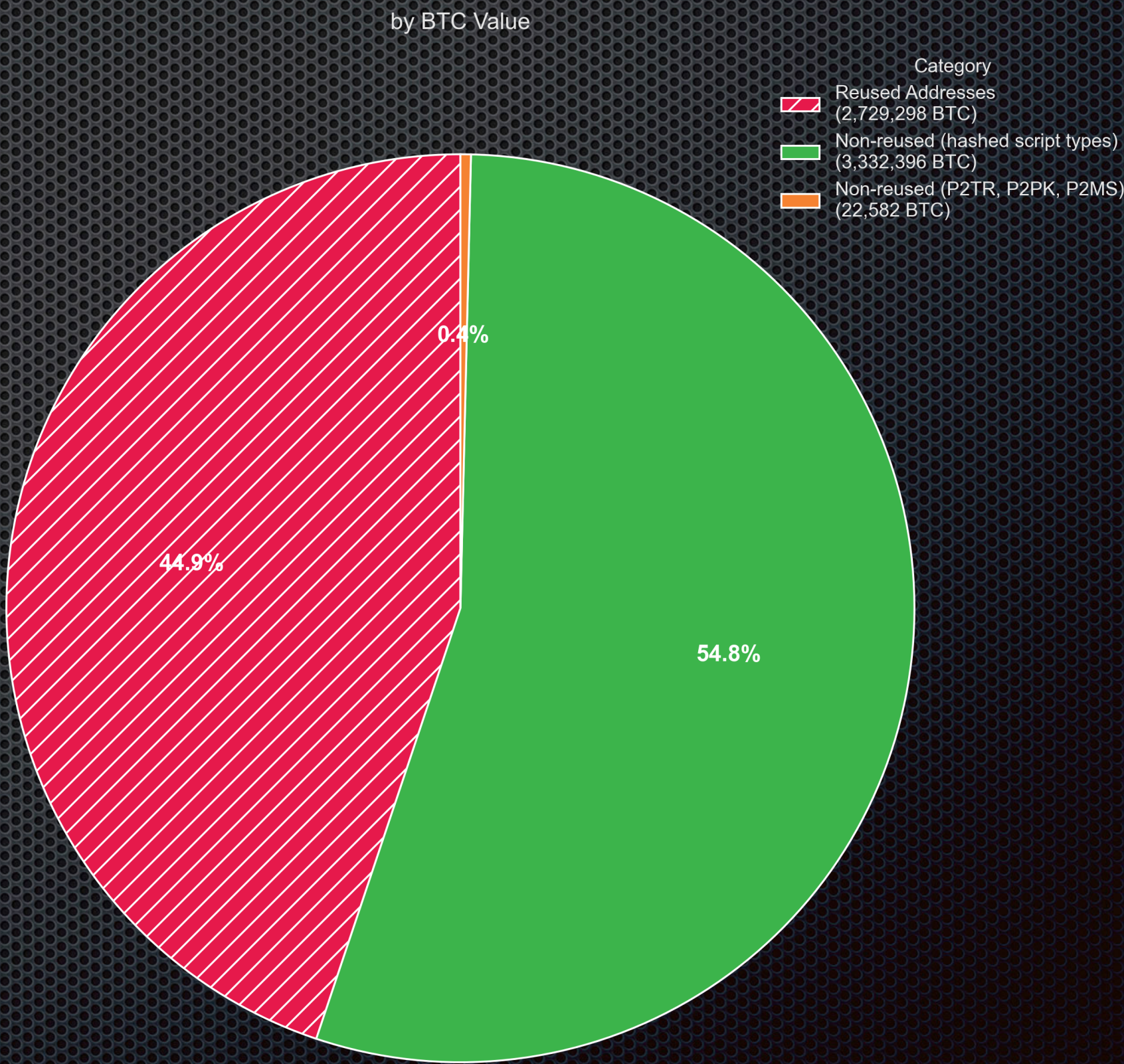
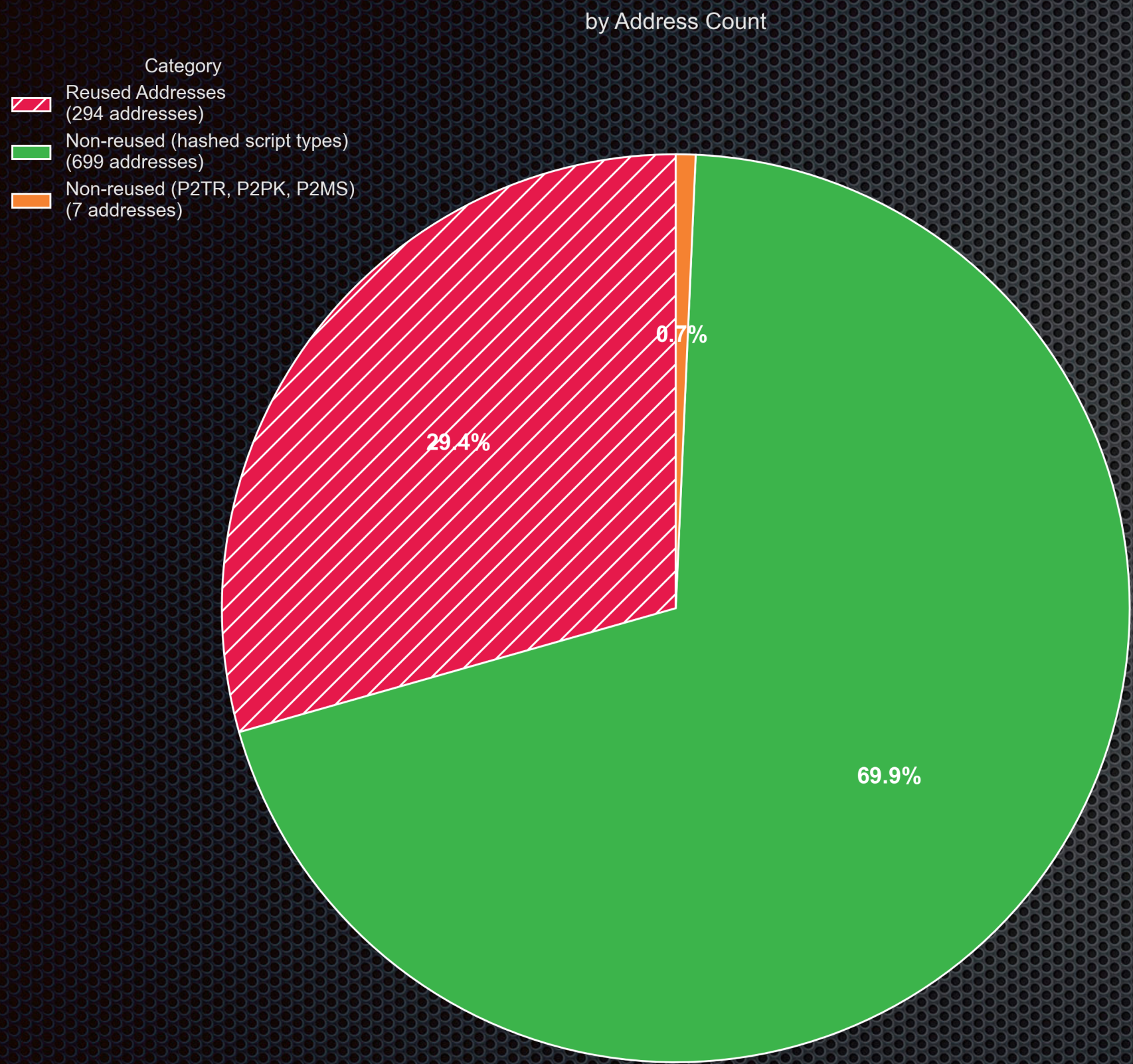
Immediately vulnerable



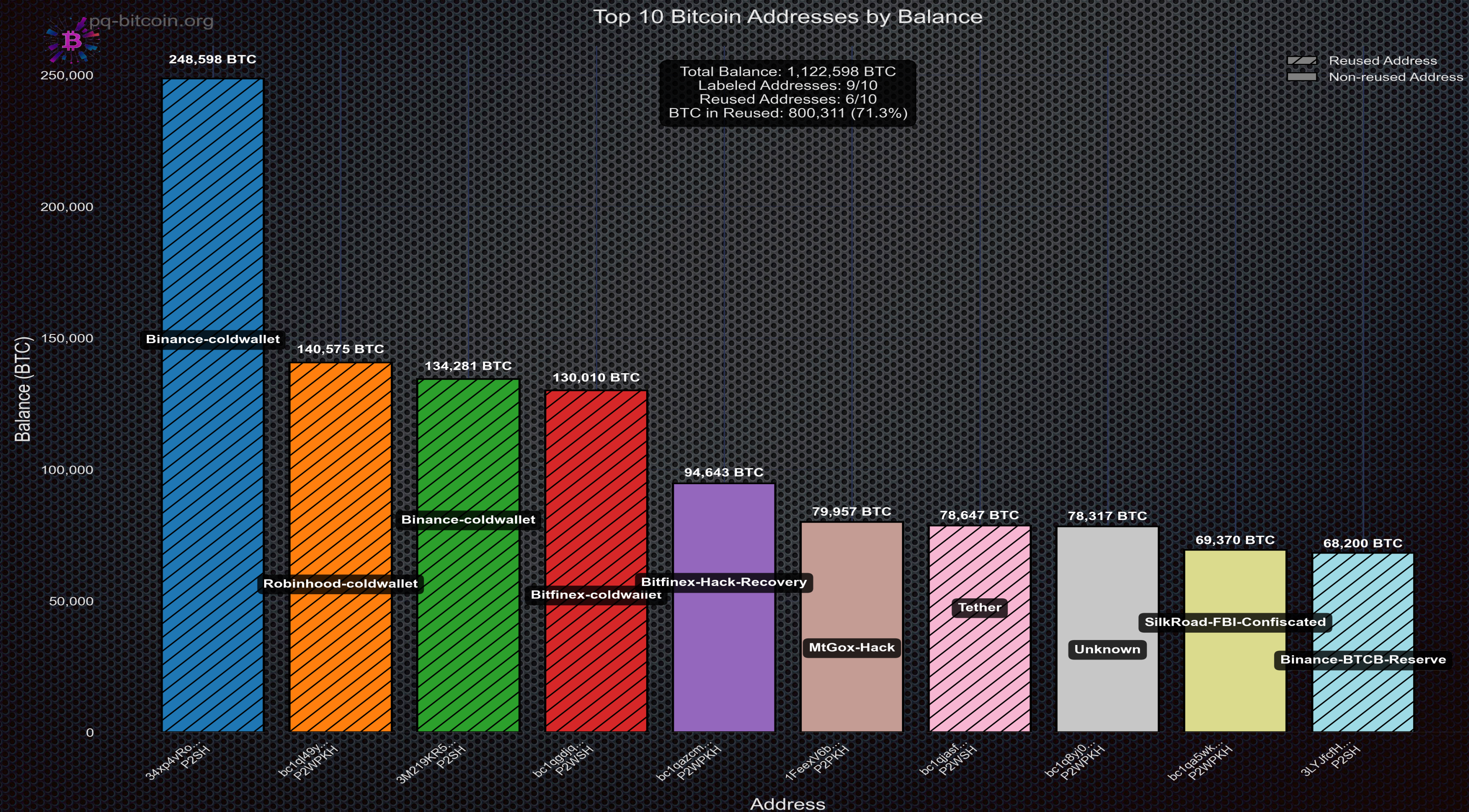
Reuse status - Top 1000

 pq-bitcoin.org

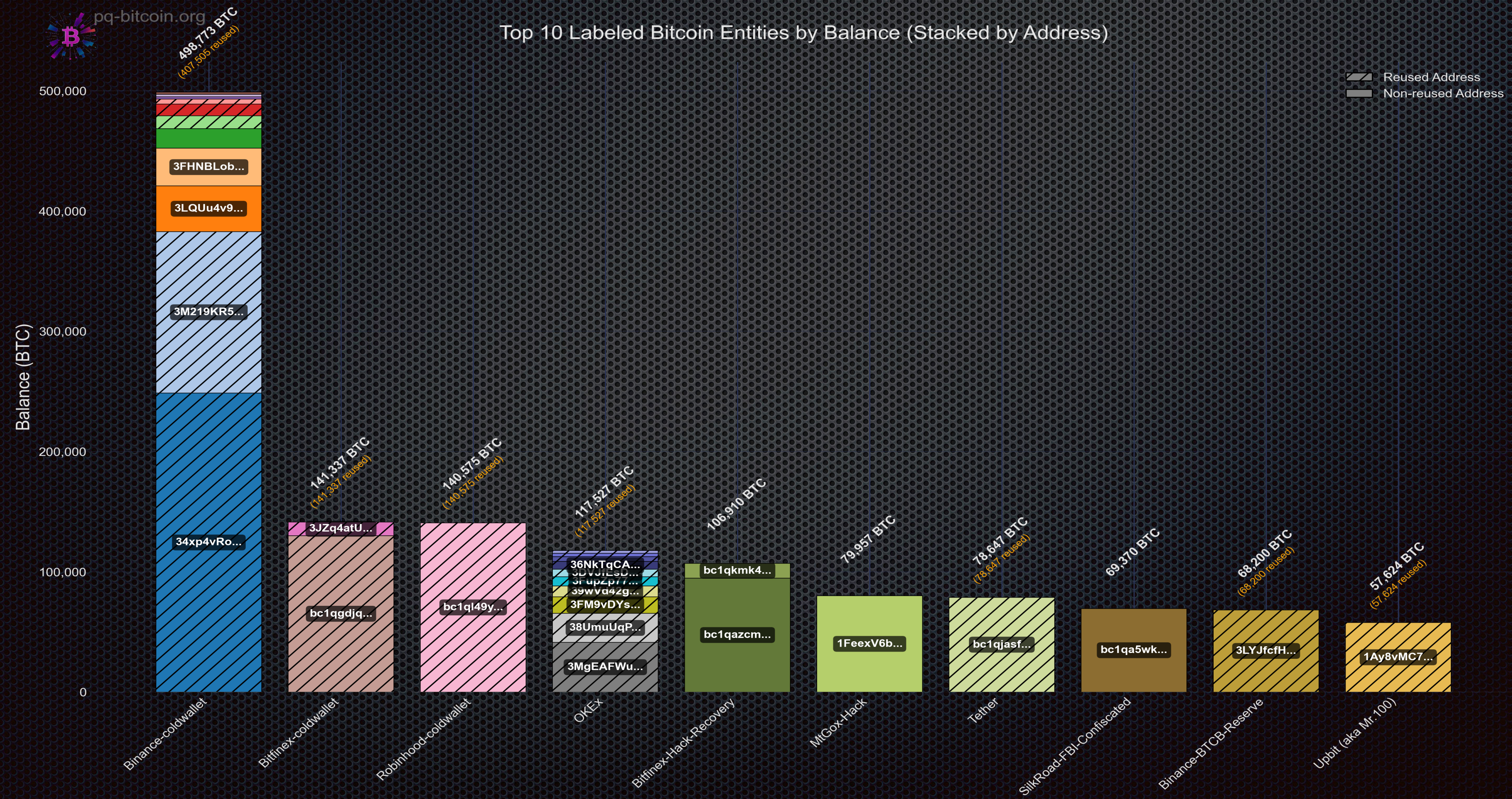
Breakdown of Address Reuse Status in Top 1000 Addresses



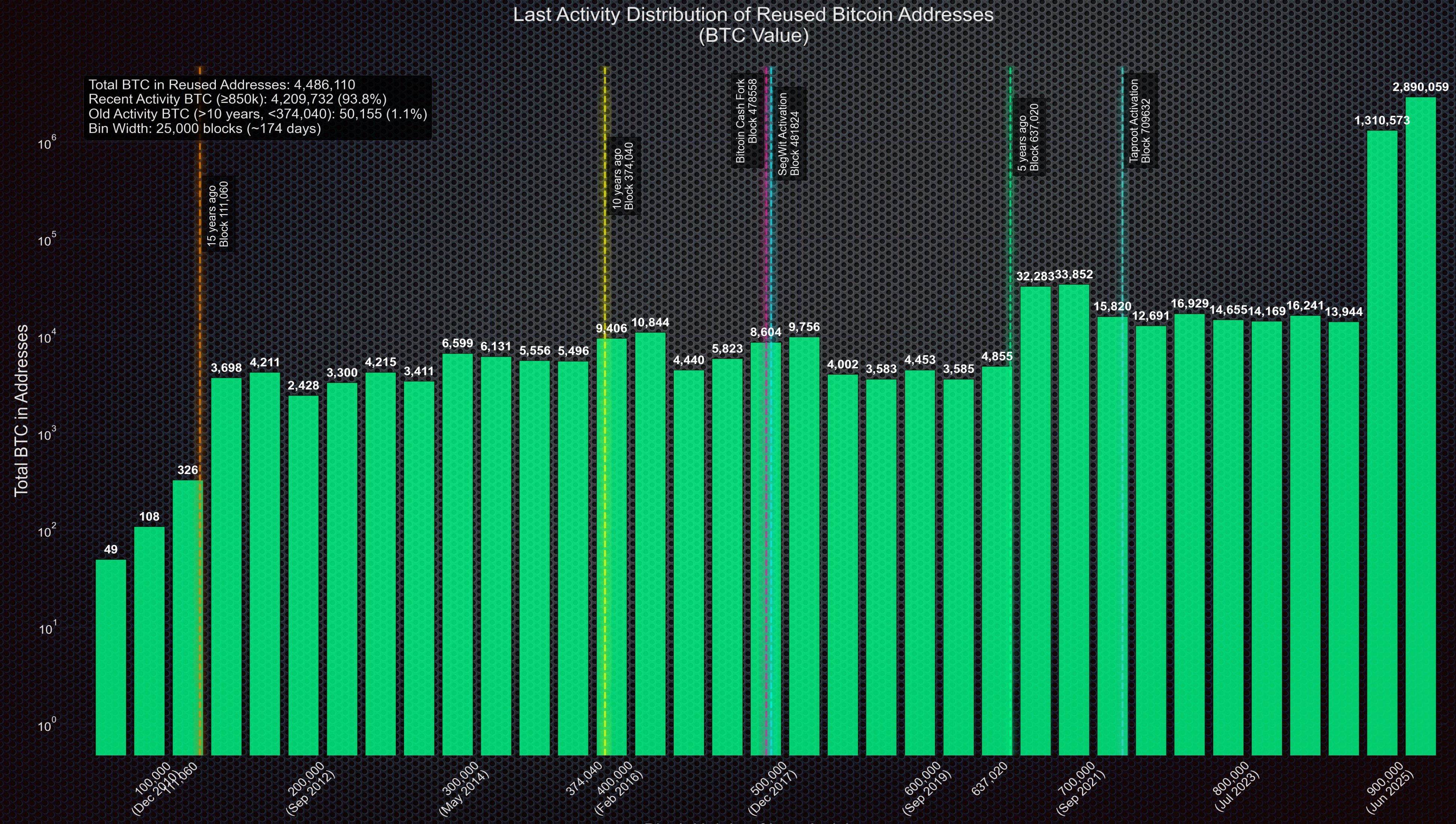
Top Addresses



Top 10 labeled



Activity time (Logarithmic scale!)

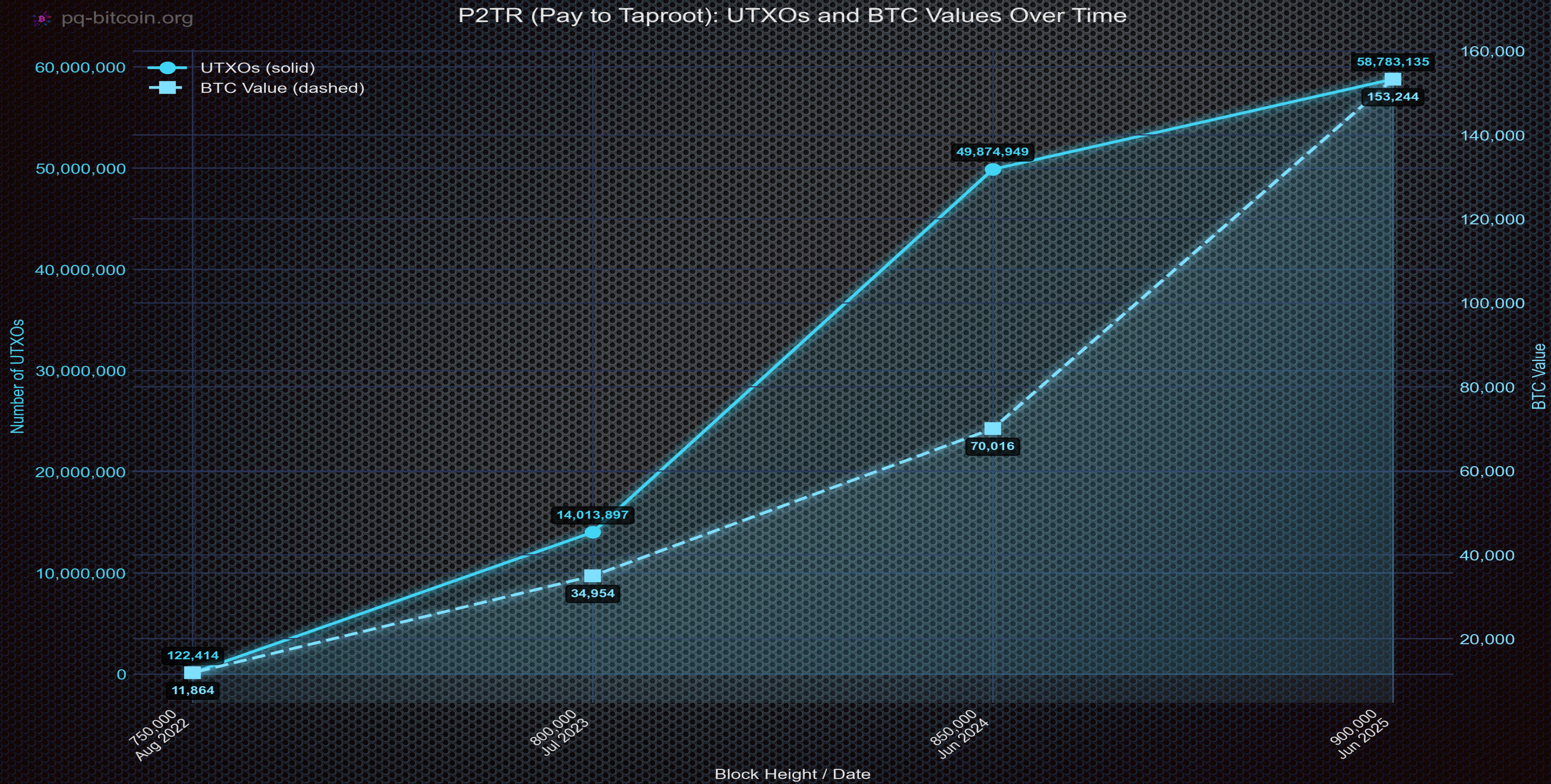


Reused addresses

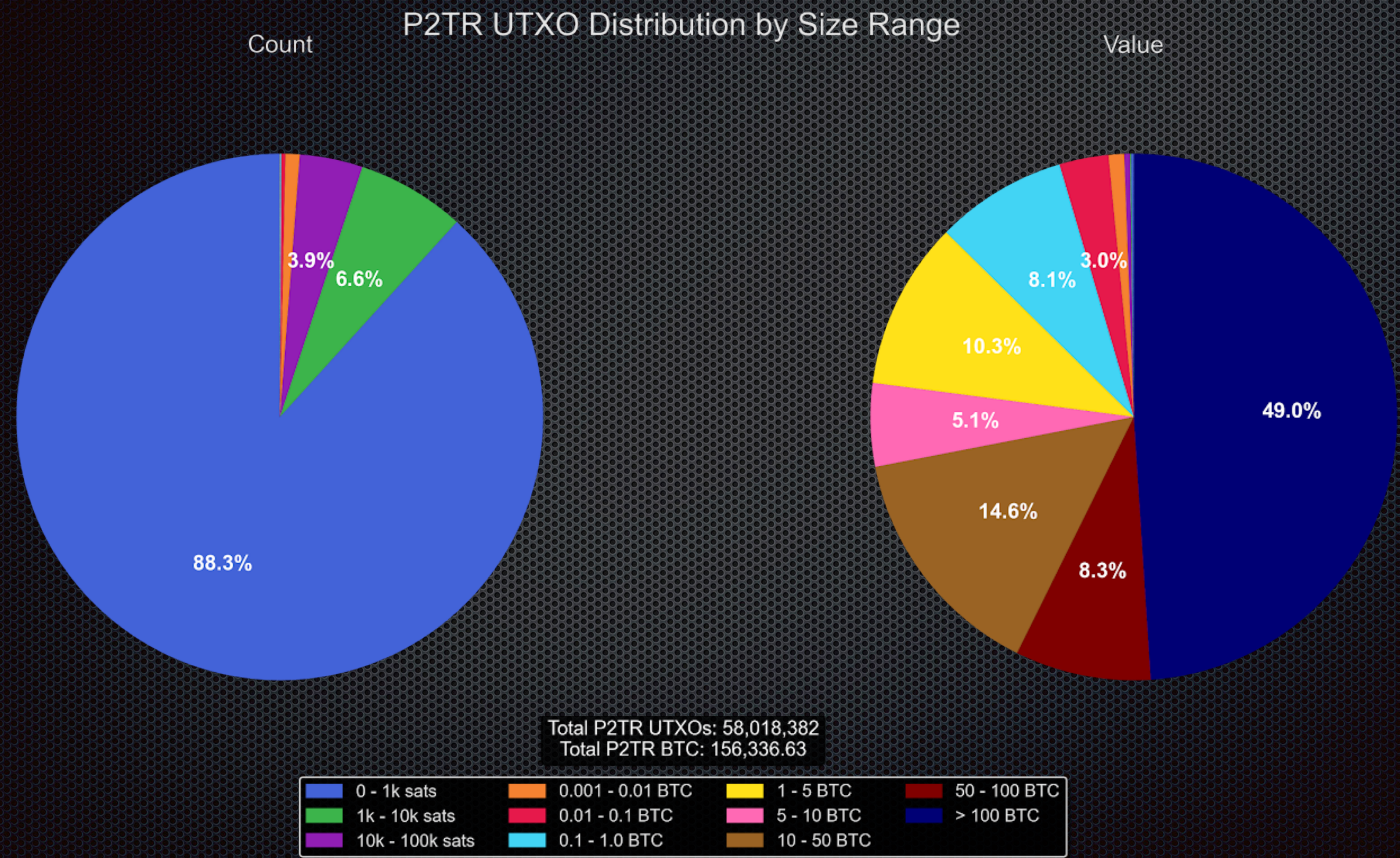
- ✦ Most of the funds have been active recently
- ✦ Significant amounts are with known entities

Activity Period	Addresses	Address %	Total BTC	BTC %
Within last 5 years	5354856	63.73	4371782.98	97.45
Between 5 and 10 years	2224323	26.47	64172.36	1.43
Between 10 and 15 years	823216	9.80	49784.88	1.11
More than 15 years	221	0.00	370.20	0.01

P2TR ~ 150K BTC



P2TR ~ 150K BTC

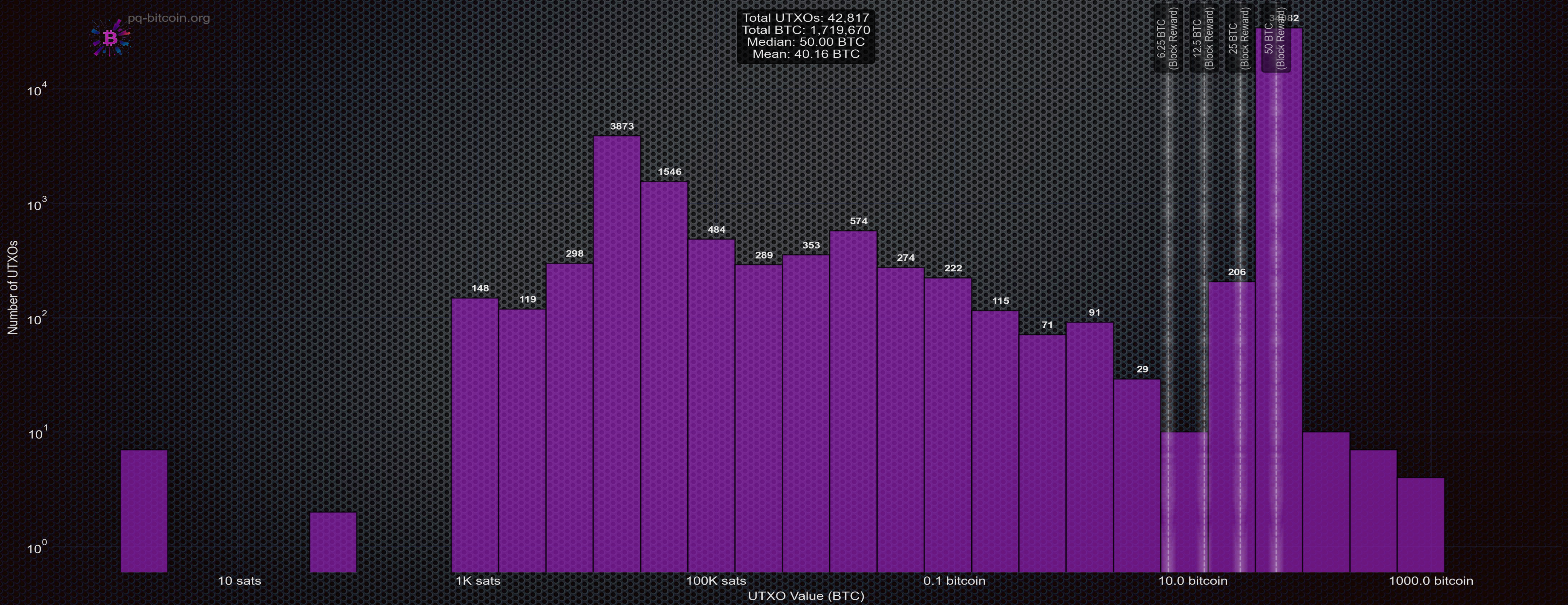


P2TR ~ 150K BTC

Size Range	UTXO Count	Count %	Value (BTC)	Value %
0 - 1k sats	51,218,283	88.3%	248.9	0.2%
1k - 10k sats	3,855,459	6.6%	114.3	0.1%
10k - 100k sats	2,242,806	3.9%	545.8	0.3%
0.001 - 0.01 BTC	505,716	0.9%	1.5k	1.0%
0.01 - 0.1 BTC	145,261	0.3%	4.7k	3.0%
0.1 - 1.0 BTC	40,064	0.1%	12.6k	8.1%
1 - 5 BTC	8,029	0.0%	16.1k	10.3%
5 - 10 BTC	1,169	0.0%	8.0k	5.1%
10 - 50 BTC	1,223	0.0%	22.9k	14.6%
50 - 100 BTC	185	0.0%	13.0k	8.3%
> 100 BTC	187	0.0%	76.6k	49.0%

P2PK ~1.72M BTC

P2PK UTXO Value Distribution



QV The story so far

- ✦ ~6.51M BTC (32.73% of total supply) is immediately QV
- ✦ Address Reuse ~4.49M BTC likely can be moved
 - ✦ T1000: account for ~2.73M BTC high rates of reuse 30% (vs. 15-20%)
- ✦ P2PK 1.72M BTC almost entirely coinbase TX, likely can't move
- ✦ P2TR 0.15M BTC many non-economic UTXOs

Immediate Protective Measures

- ✦ **DO NOT REUSE ADDRESSES!**
- ✦ Large amounts held in vulnerable addresses can be moved
 - ✦ P2PKH, P2SH, P2WPKH, or P2WSH
- ✦ For old coins - check if addresses reuse occurred on forks
- ✦ Don't hold funds in exchanges (or verify how they manage key security)

Migration Pathways

- ✦ Migration of all utxo can take between 3 months and 2 years or more
- ✦ There are several mechanisms, some permissive and some protective:
 - ✦ CDR - Commit, Delay, Reveal
 - ✦ Hourglass strategy - Rate limit vulnerable utxo spend
 - ✦ QRAMP - Enforcing mandatory migration period

Burn vs. Steal Dilemma

- ✦ A **CRQC** is a Security Bug
- ✦ What do we do with coins that did **not migrate**?
- ✦ **Burn** - coins can never move
 - ✦ Philosophical clash
- ✦ **Steal** - Assets claimed by those with a CRQC
 - ✦ Destabilize the ecosystem
- ✦ **Hourglass** - rate limit

Timeline for upgrade

Three main phases:

1. Research

2. Implementation

3. Migration

Long-Term

2.5 Y

1.5 Y

3 Y

~7 Y

Contingency

0.5 Y

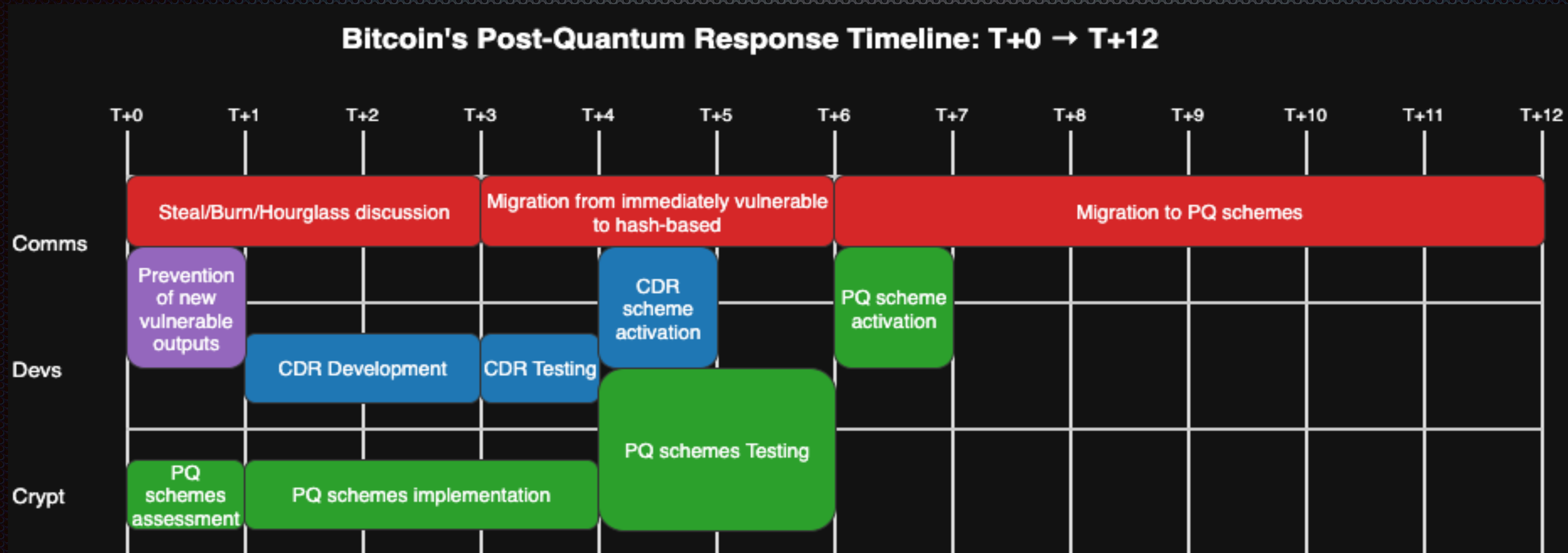
1 Y

1 Y

~2 Y

Emergency play book

- 3 teams: Comms, Devs, Crypt



Ecosystem Vulnerabilities

- ✦ Bitcoins lives on the [internet](#) (SSL, DNS, etc.)
- ✦ [P2P](#)-level attacks
- ✦ Signatures on [Core versions](#)

Bitcoin & Quantum Computing

- ✦ Bitcoin threat model
 - ✦ Mining
 - ✦ Signatures
- ✦ Current state of the utxos
- ✦ Timeline to solution



For more details:

pq-bitcoin.org



PQ report:

[chaincode.com/bitcoin-
post-quantum.pdf](https://chaincode.com/bitcoin-post-quantum.pdf)

